COMICCRAFT — SCALABILITY & FUTURE PLAN

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CURRENT LIMITATIONS

- Runs only on localhost
- Fixed 5-panel structure
- English-only text generation
- Images are generated sequentially (not parallel)
- No user accounts or saved comics

SCALABILITY ROADMAP

Phase 1 (Next 1 month)

- Deploy backend to Render or Railway (free tier)
- Deploy frontend to Streamlit Cloud or Vercel
- Add custom panel count (3, 5, or 8 panels)
- Parallel image generation using asyncio.gather

Phase 2 (3 months)

- User accounts with saved comic history
- Character consistency across panels (using reference images)
- Multi-language support (Hindi, Tamil, Spanish)
- Custom fonts and page layouts for PDF

Phase 3 (6 months)

- Voice narration for comics (AI text-to-speech)
- Video export (MP4 with timed panel transitions)
- Integration with social media for direct sharing
- Mobile app using React Native or Flutter

Phase 4 (1 year)
- Collaborative story creation (multiple users)
- AI-generated sound effects and background music
- Comic series (multi-episode) support
- Marketplace for sharing comics publicly

TECHNICAL SCALABILITY

- FastAPI handles 1000+ concurrent requests with Gunicorn workers
- Pollinations.ai scales automatically
- Gemini API scales with paid tier (production)
- Static image serving via CDN for faster loading
- Database integration (PostgreSQL) for user data

MONETIZATION (OPTIONAL)

- Free tier: 3 comics per day
- Pro tier: Rs. 199/month for unlimited + HD images
- Creator tier: Rs. 499/month with premium AI styles
- Enterprise: API access for schools and media companies

PERFORMANCE IMPROVEMENTS PLANNED

- Use async HTTP client (httpx) instead of requests
- Cache frequently used prompts
- Pre-generate popular image styles
- Reduce image size for faster loading (512x512 preview)